# Emergence of Cooperation in Group Interactions: Avoidance vs Restriction

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# Outline

1. Introduction
   - The problem of the evolution of cooperation
   - Cooperation in Public Goods Game
2. Two models of commitment in Public Goods Game
   - Restriction and avoidance
   - Levels of participation

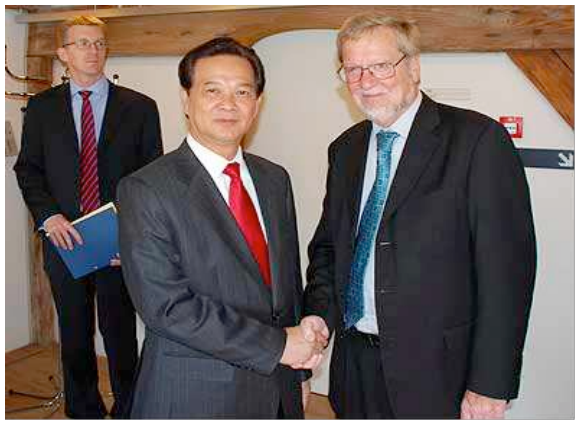
# What is Cooperation?

- *pay a cost for someone else to receive a benefit.*
- Cooperation is widespread: insects, hunter-gatherer societies, team work, international relationships, etc.



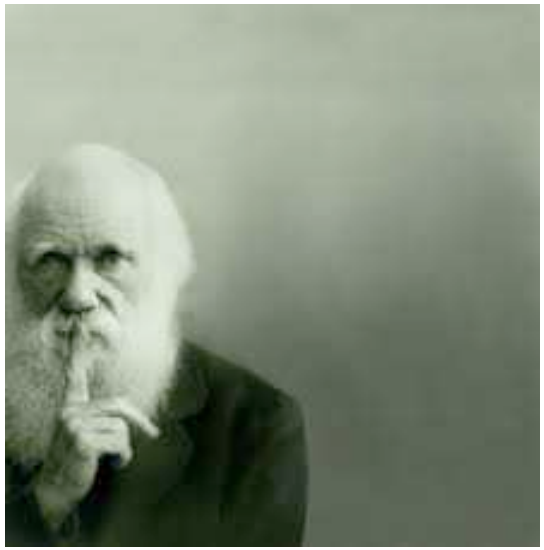






# Paradox of cooperation

- Natural selection: *only the fittest survive*
  - Everyone wants to increase their fitness.
  - No one wants to pay the cost, happily accepting benefits.
- *Dilemma*: everyone would be better off cooperating with each other (*benefit* $>$ *cost*).



IF NATURAL SELECTION IS BASED ON COMPETITION, HOW CAN IT LEAD TO COOPERATION?

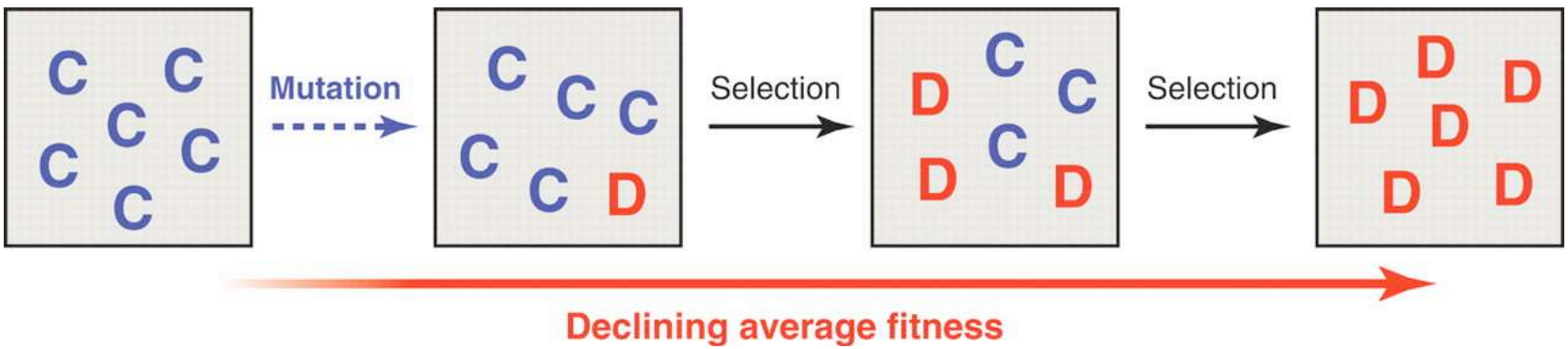
# Interdisciplinary research

- Understanding the evolution of cooperation remains a **fundamental challenge**, for scientists from fields like evolutionary biology, physics, economics, mathematics, computer science[1], etc.
- Many fields ... same math!!
- **Mathematical Framework**: Evolutionary Game Theory
- **Metaphors**: Prisoner's dilemma, Public Goods Game, Ultimatum game, etc.

[1] IJCAI, AAMAS, CEC, GECCO, ALIFE, COGNITION

# Evolutionary Game Theory

payoff $\longrightarrow$ fitness $\longrightarrow$ social success




Natural selection leads to the destruction of cooperation (**tragedy of commons**).

# Why arrange commitments?

- Sometimes **difficult to predict others' behavior or recognize their intentions** with sufficient confidence.

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- Sometimes **difficult to predict others' behavior or recognize their intentions** with sufficient confidence.
- *Commitment proposal* can help clarify intentions of others.
  - contracts, marriage, apartment rental, etc.





# From pairwise [2] [3] [4] to group commitment





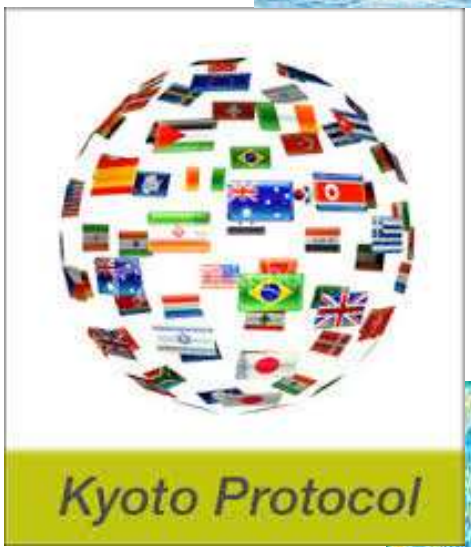
[2] Han, Pereira, Santos. **Emergence of commitment and cooperation**. AAMAS, 2012.

[3] Han, Pereira, Santos, Lenaerts. **Good agreements make good friends**. Nature Scientific Reports, 2013.

[4] Han, Santos, Lenaerts, Pereira. **Synergy between intention recognition and commitments in cooperation dilemmas**. Nature Scientific Reports, 2015.

# Group commitment applications





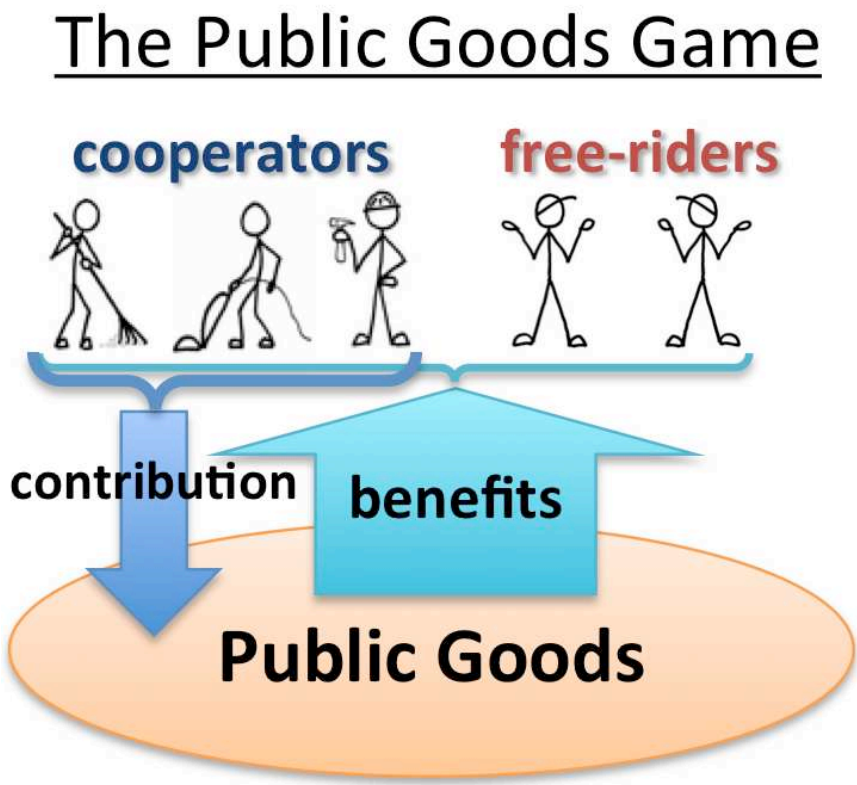







# The Public Goods Game (PGG)

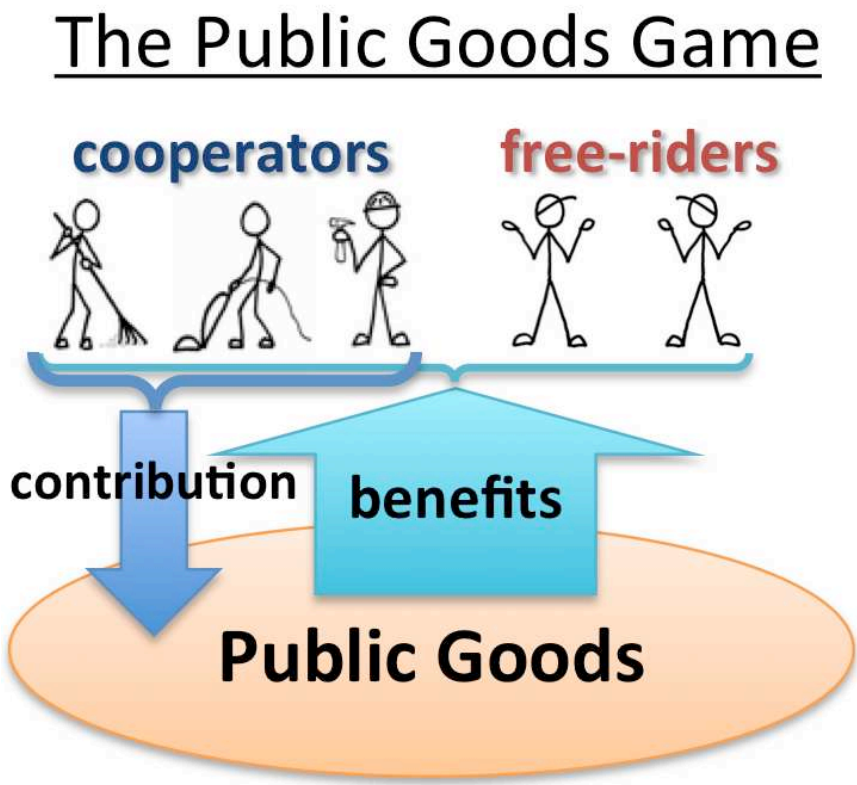
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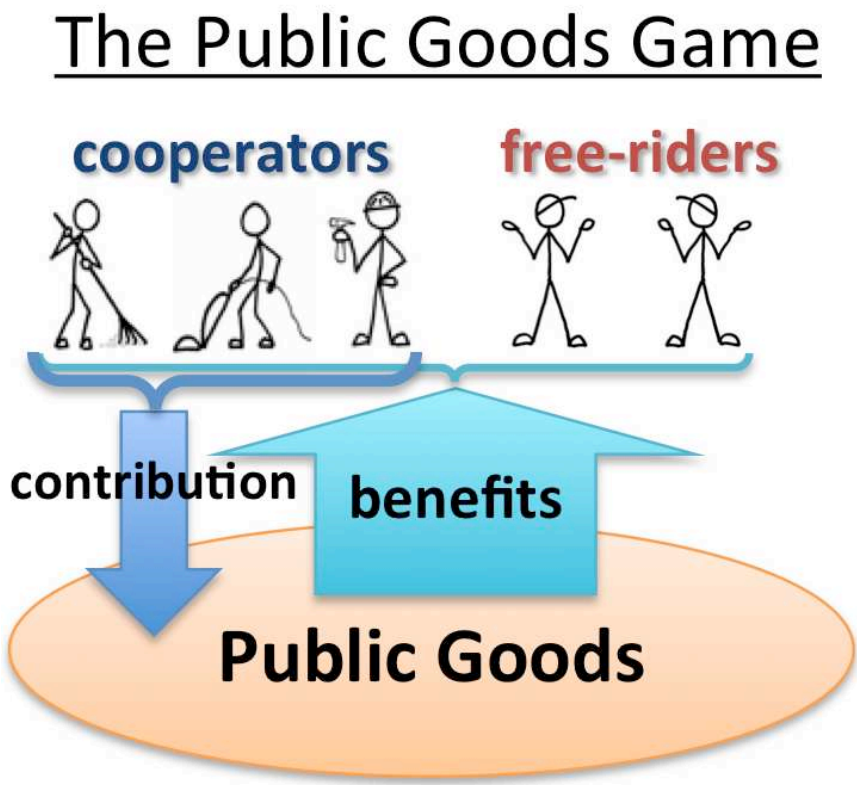


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- Free-rider's payoff: $\frac{r \times N_c \times c}{N}$
- Cooperator's payoff: $\frac{r \times N_c \times c}{N} - c$



# Restriction vs Avoidance of non-committers

- What to do when other players do not commit to the group effort?
- Refusing to participate removes benefits from free-riders, but also from those who are willing to join and contribute.
- Restricting access of non-committers to public goods needs extra effort.

# Commitment proposing strategies

**AVOID** contributes if everyone in the group commit, otherwise does not contribute.

**RESTRICT** sets up extra measure to restrict access of non-committers to the PGG.



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- commitment proposers share a set-up cost, $\epsilon_P$.
- compensation from dishonored committers, $\delta$.
- extra measure set-up cost, $\epsilon_R$
- restriction for non-committers, $\Psi < 1$



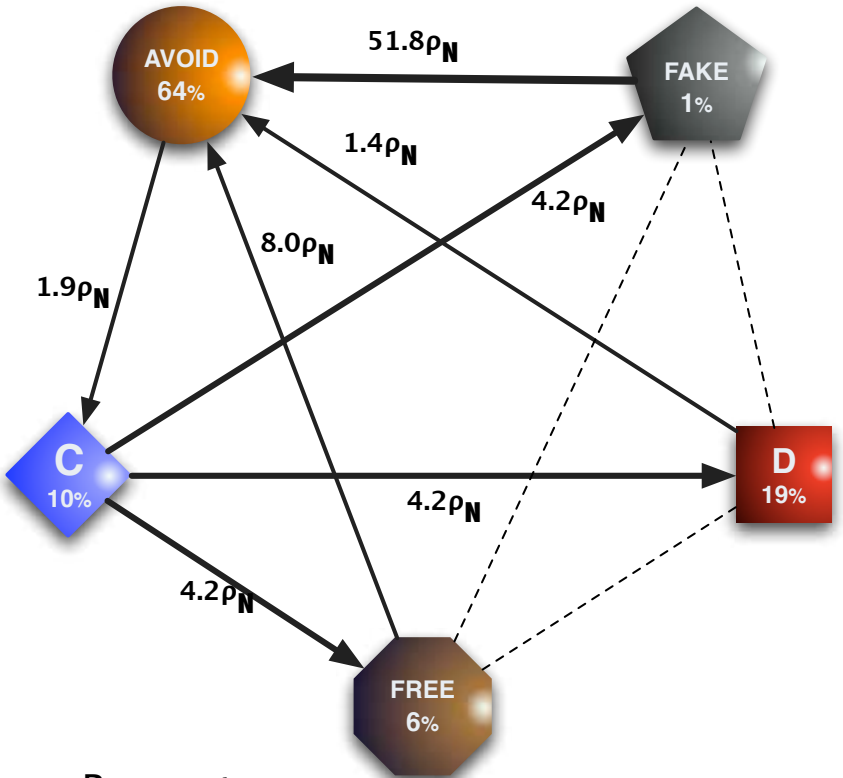
# What are the strategies of the co-players

- Cooperator (C): always accepts commitment.
- *Defector (D)*: never accepts commitment.
- *Fake committer (FAKE)*: accepts commitment, yet defects in game.



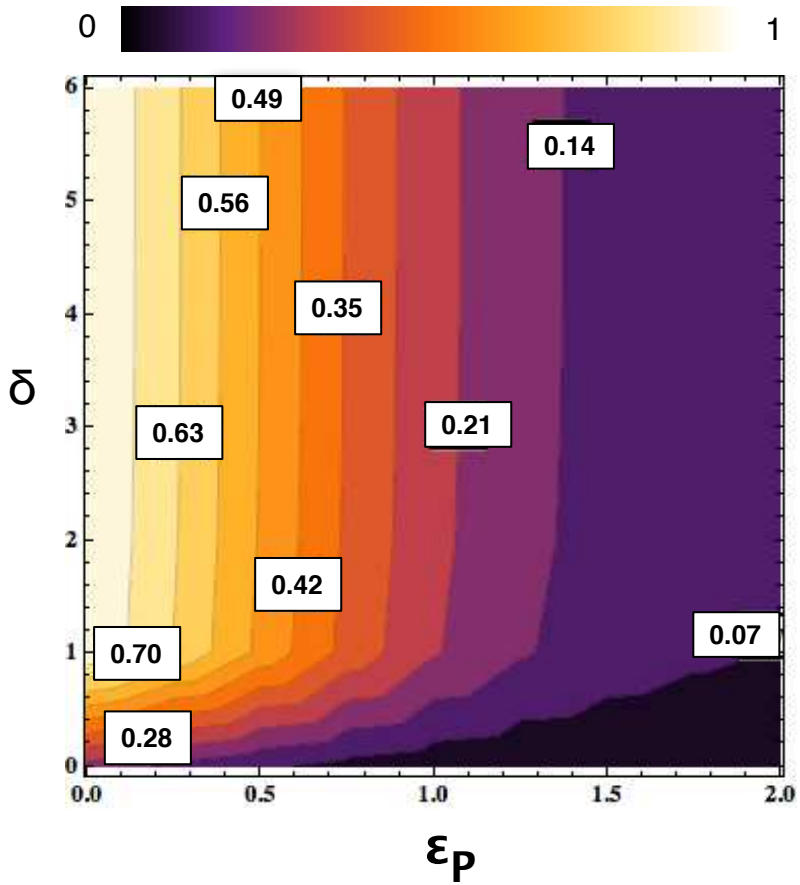

- *Commitment free-rider (FREE)*: accepts commitment and cooperates, yet defects when receiving no commitment proposal.

# AVOID non-committers promotes cooperation

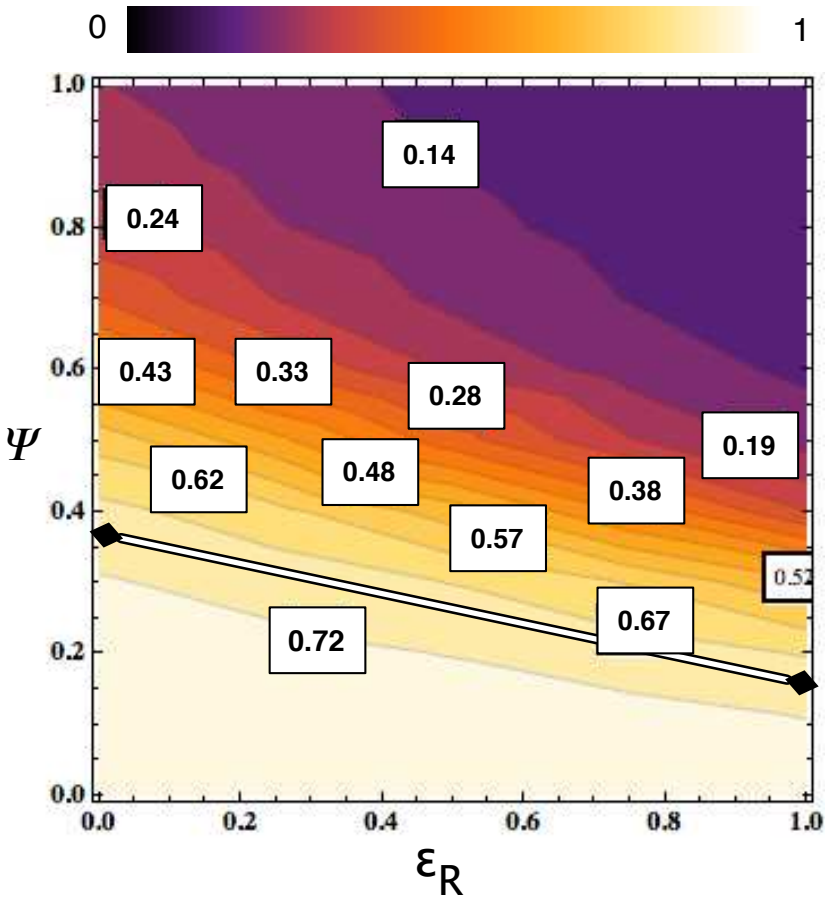



Parameters:

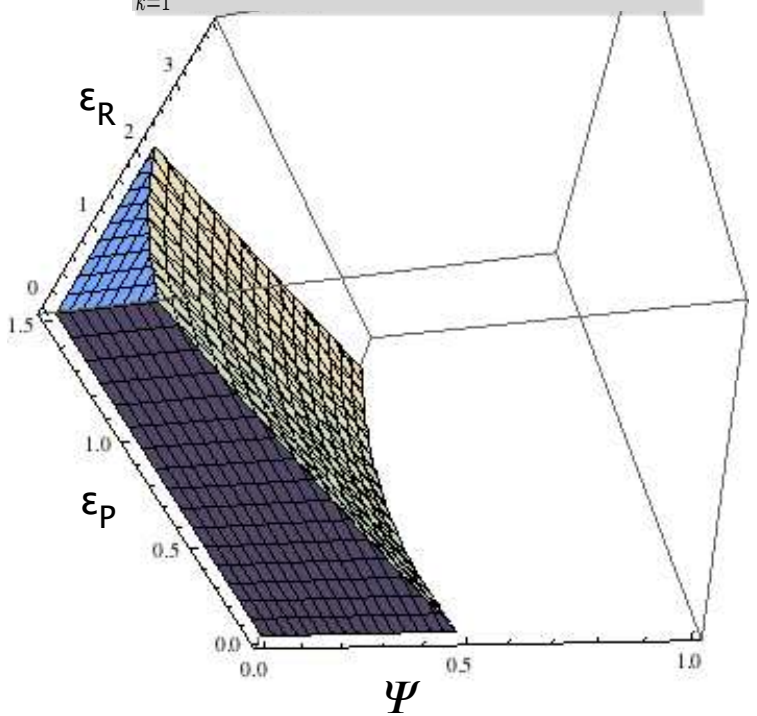
$\varepsilon_P$=0.25; δ=2; r = 3; N = 5

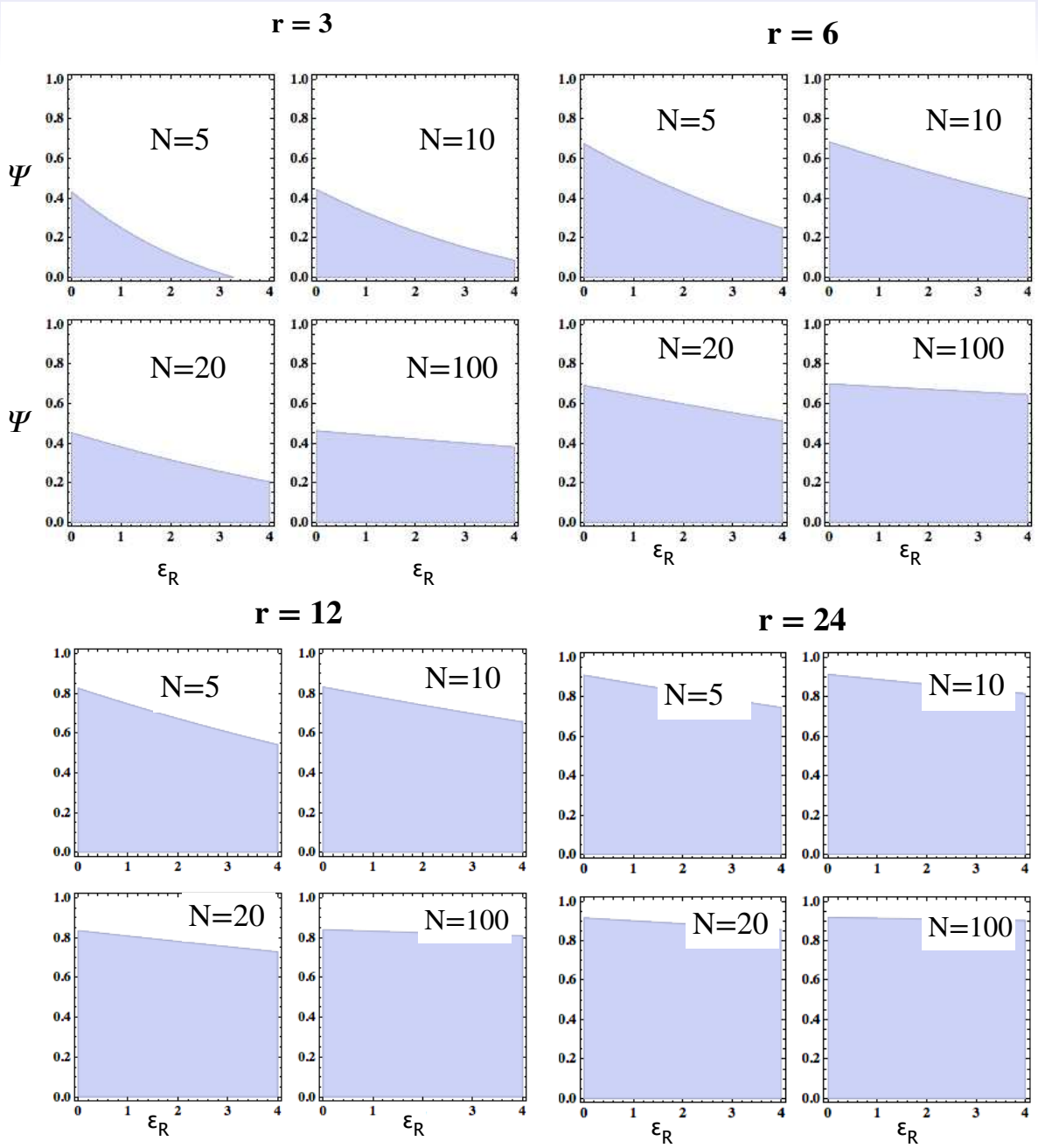


# ... but RESTRICT can do better




$$\sum_{k=1}^{N-1} \frac{k(1-\psi)}{\psi N + k(1-\psi)} rc \geq F_N \epsilon_R + F_{N-1}\epsilon_P + (N-1)c$$




r = 3
r = 6
r = 12
r = 24
N=5
N=10
N=20
N=100
Ψ
Ψ
εR
εR
εR
εR
εR
εR
εR
εR

# Remark [5].

- Both commitment strategies can promote evolution of cooperation in PGG when the cost of arranging commitment is justified with respect to the benefit of cooperation.
- RESTRICT is better than AVOID if non-committers can be efficiently restricted, especially for beneficial PGG.

[5] Han, Pereira, Lenaerts. **Avoiding or restricting defectors in public goods games?** Royal Soc Interface, 2015.

# Group level of participation

- How many participants need to commit to start the venture?
- Multiple intermediate degrees of group commitment are possible.
- Minimum membership requirement is standard in international agreement (*Montreal protocol – 11 countries 1989; Kyoto Protocol – 55 parties 2005; Paris agreement: expected 85*)

# Commitment strategy

**COMP(F)** contributes if there are at least $F$ players in the group commit, otherwise does not contribute.

There are as many commitment proposing strategies as the group size: $1 \leq F \leq N$.



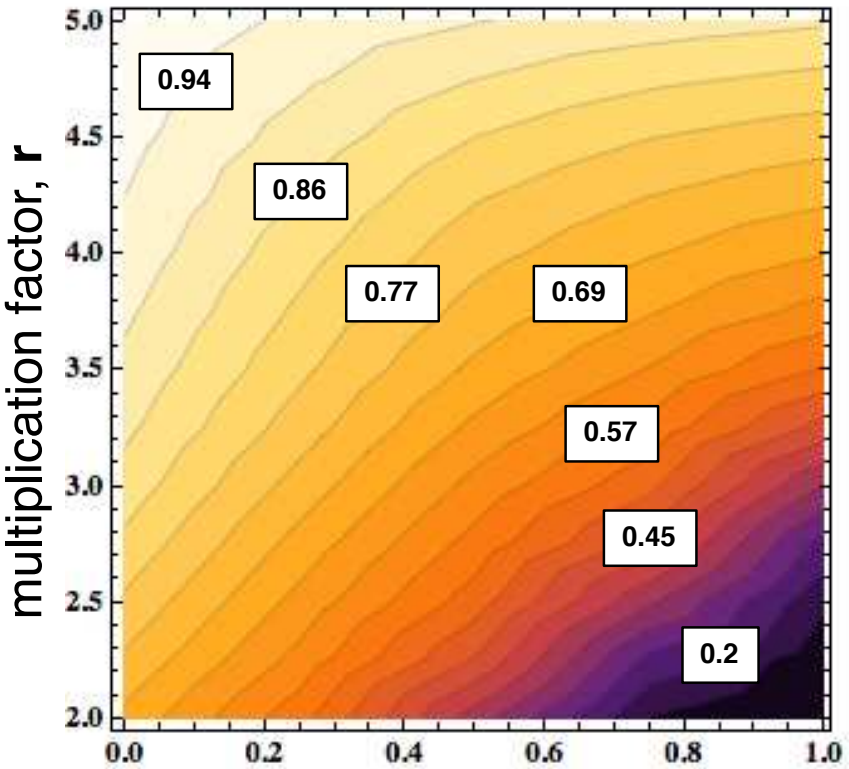
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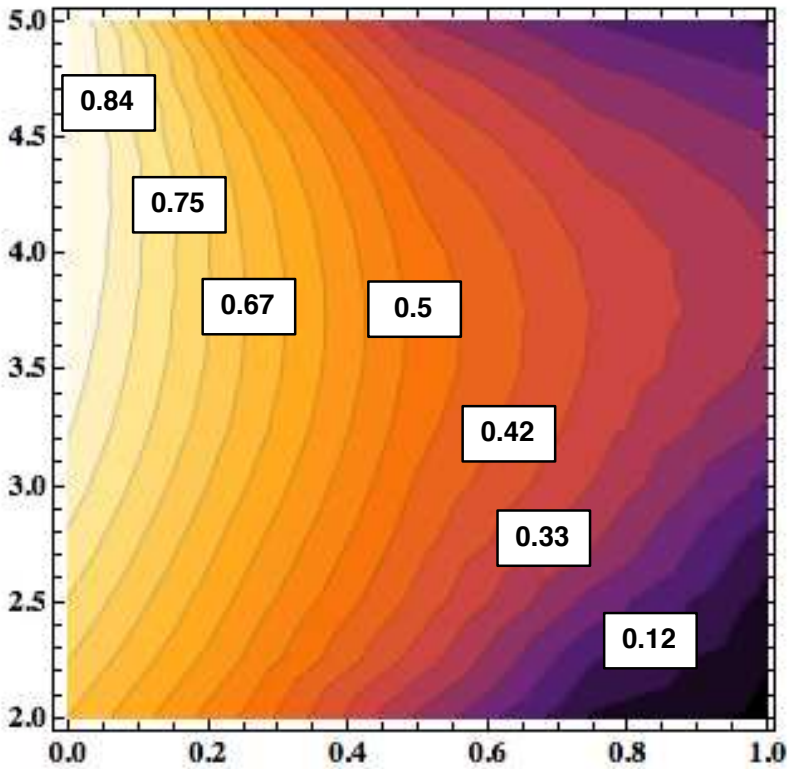

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# Emergence of cooperation and commitment

Cooperation frequency

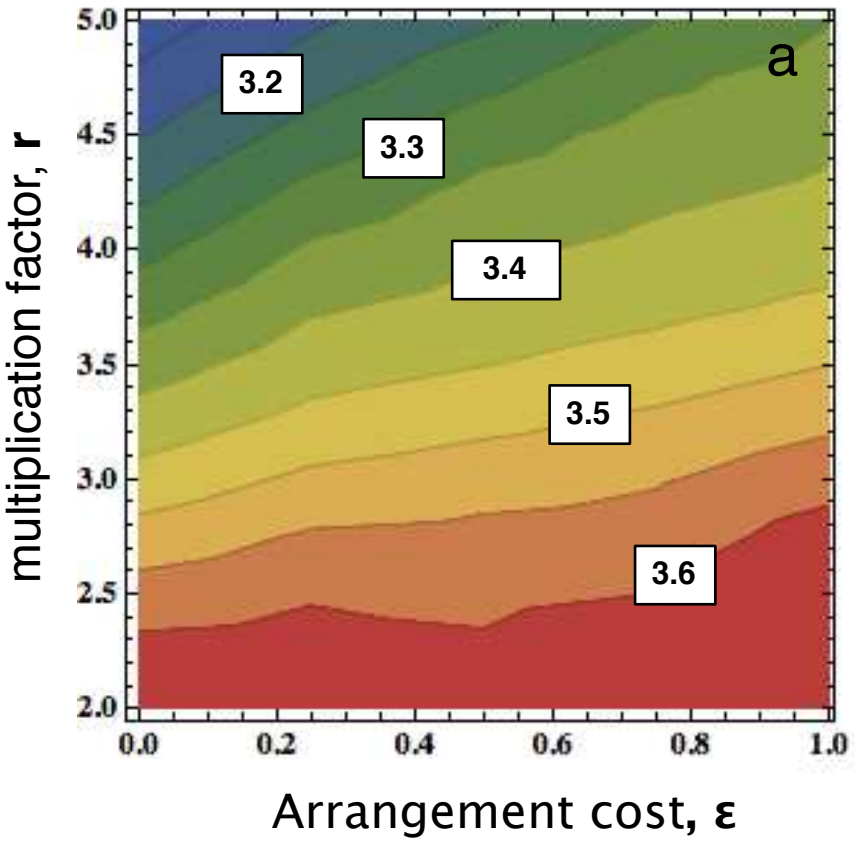



Commitment frequency

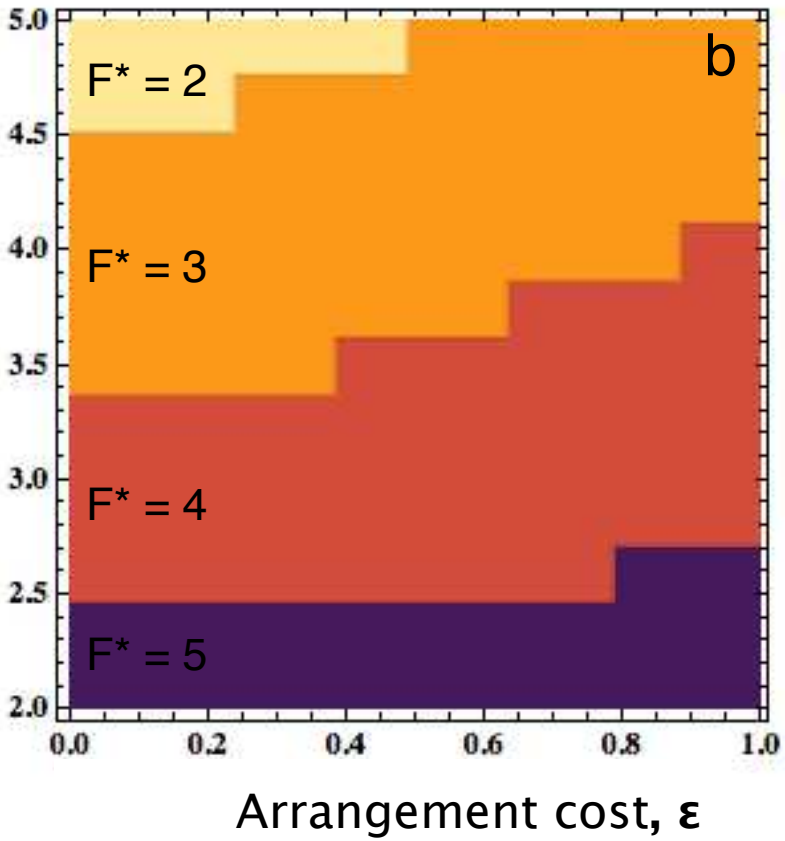


# How many members need to commit?

Average commitment level




Optimal commitment strategy



# Public Goods Game Experiments with Commitments

- Prior-deposit model [Cherry and McEvoy, 2013]:
  - Centralized party arranging the commitment.
  - Commitment threshold $F$ is decided externally.
  - Best to ask for full group commitment ($F = N$) (experiments).
  - Our model is modified to capture this experiment setting and results comply with this observation.

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- Forced cooperation model [Chen and Komorita, 1994]: once agreed to cooperate, forced to cooperate.
  - FAKE is equivalent to FREE.
  - Four strategies model: 60% cooperation, similar to data.

# Remarks [6]

- Conclusions for pairwise commitments can be generalized to group commitments (regarding arrangement cost and compensation).
- In multi-player games, an intermediate commitment is emerged.
- The more beneficial the cooperation and the lower cost of arranging commitment, a lower degree of commitment from other members will be required.
- Complying with behavioral experimental data.

[6] Han, Pereira, Lenaerts. **Evolution of Commitment and Level of Participation in Public Goods Games.** In Revision for JAAMAS.

# Conclusions

- Commitment strategies can promote cooperation if its arrangement is well facilitated.
- Restricting non-committers' access to the public goods is more advantageous if it is facilitated (small cost, high effect).
- Otherwise, it is important to look at the level of participation depending on different factors.

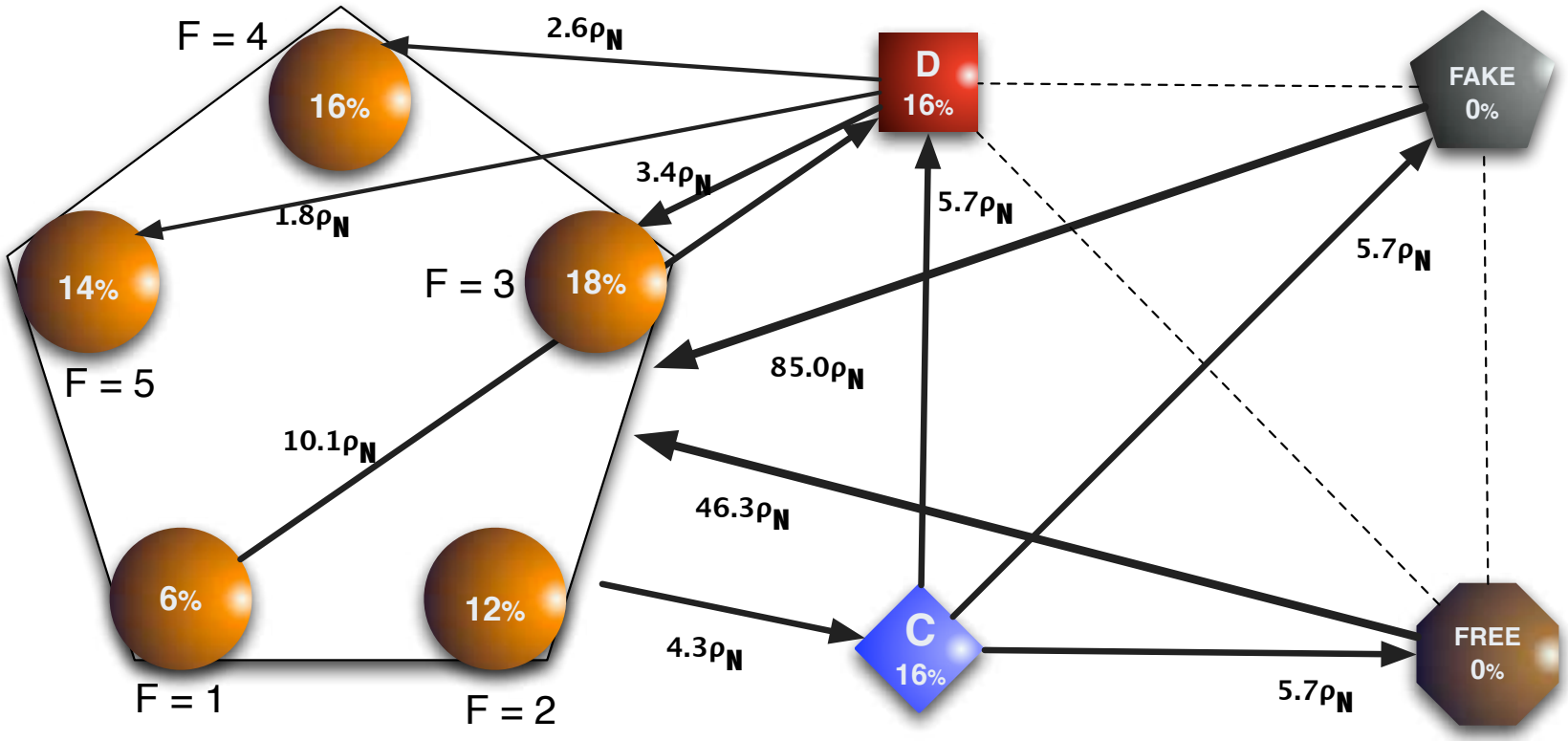
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- **Implications**: insights into design of self-organized and distributed MAS where agents work in team (e.g. multi-robot systems).

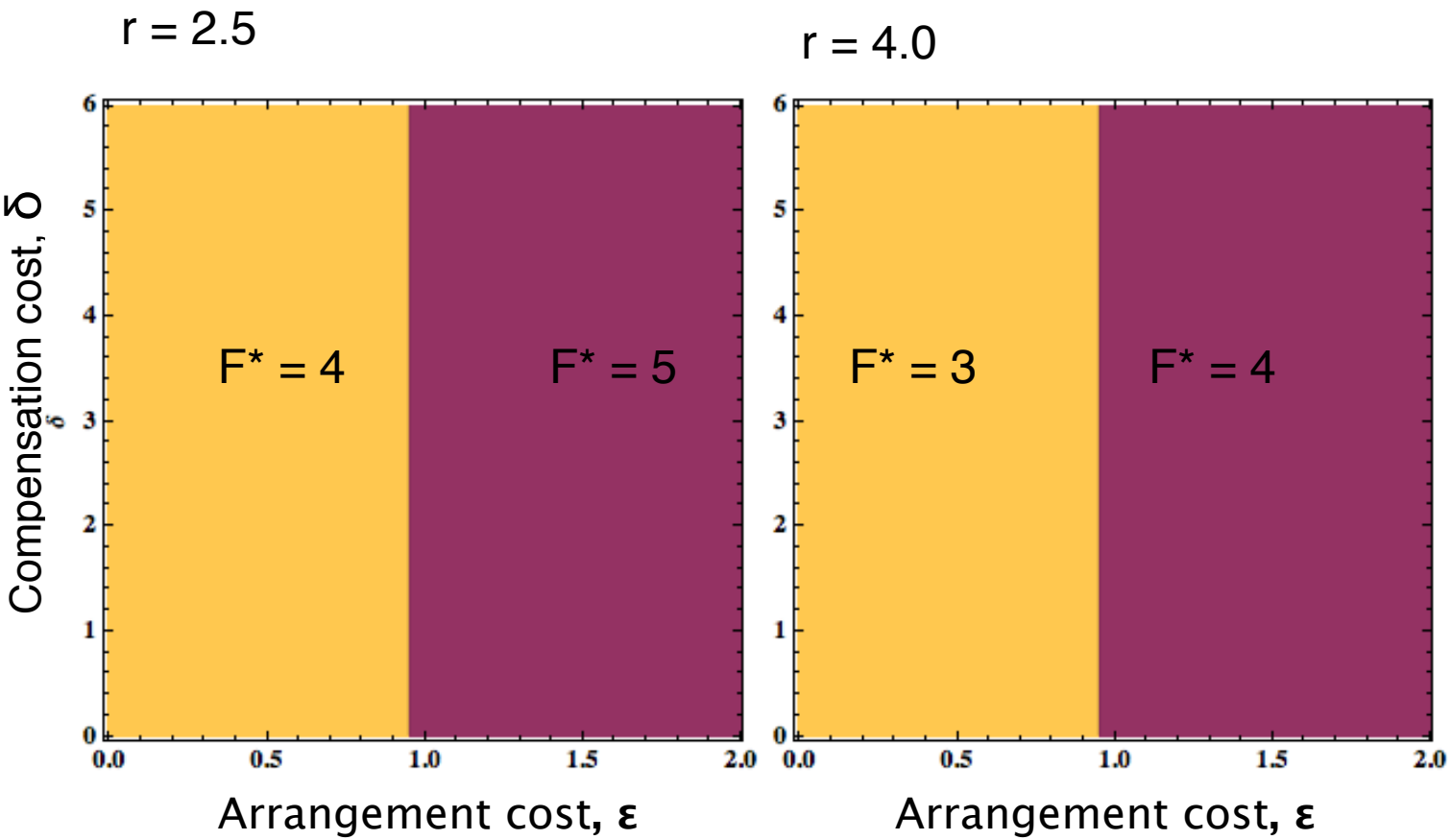
# Thank you!

QUESTIONS?

# Viability of COMP(F)



# Optimal F for varying $\epsilon$ and $\delta$



# Varying group sizes

